

NUMERICAL PROBLEMS IN WAVE OPTICS

MANDATORY PROBLEMS

1. How does the fringe width of interference fringes change, when the whole apparatus of Young's experiment is kept in a liquid of refractive index, 1.33?
2. Laser light of wavelength 630 nm incident on a pair of slits produces an interference pattern in which the bright fringes are separated by 7.2 mm. Calculate the wavelength of another source of laser light which produce interference fringes separated by 8.1 mm using same pair of slits.
3. In Young's double slit experiment, the two slits 0.15 mm apart are illuminated by monochromatic light of wavelength 450 nm. The screen is 1.0 m away from the slits. (i) Find the distance of the second
 - (a) bright fringe,
 - (b) dark fringe from the central maximum.(ii) How will the fringe pattern change if the screen is moved away from the slits?
4. In Young's double slit experiment, interference fringes are observed on a screen a distance kept at D from the slits. If the screen is moved towards the slits by 5×10^{-2} m, the change in fringe width is found to be 3×10^{-5} m. If the separation between the slits is 10^{-3} m, then, calculate the wavelength of the light used.
6. Light from a narrow slit passes through 2 parallel slits, 0.2 cm apart. Find the wavelength of light, if their interference fringes on a screen 1m distant from the 2 slits are 0.3 mm apart. If the whole apparatus is immersed in water of RI $4/3$, find the change in the fringe width.
7. In a fraunhofer diffraction at a single slit, when a light of wavelength 589.3 nm is used, the distance between the first minima on either side of central maximum is 0.028 m. The distance of the screen is 1 m from the plane of the slits. Calculate the width of the slit.
8. A convex lens of focal length 0.4 m forms a Fraunhofer diffraction pattern of a slit 0.02 mm wide. Calculate the distance of the 3rd bright band from the central maximum, if the wavelength of light used is 690 nm.
9. An oil immersion microscope can just resolve 2 point objects separated by 0.5 μ m, when light of wavelength 690 nm is used for illumination. If the semi vertical angle is 30° , calculate the RI of the oil.
6. If the angle between the axis of polarizer and the analyzer is 45° ; write the ratio of the intensities of original light and the transmitted light after passing through the analyzer.
7. A parallel beam of light of 500 nm falls on a narrow slit and the resulting diffraction pattern is observed on a screen 1 m away. It is observed that the first

minimum is at a distance of 2.5 mm from the centre of the screen. Calculate the width of the slit.

8. A parallel beam of light of wavelength 500 nm falls on a narrow slit and the resulting diffraction pattern is observed on a screen 1 m away. It is observed that the first minimum is at a distance of 2.5 mm away from the centre. Find the width of the slit.

9. What should be the width of each slit to obtain 10 maxima of the double slit pattern within the central maximum of the single slit pattern, for green light of wavelength 500 nm, if the separation between two slits is 1 mm?

10. In a double slit experiment using light of wavelength 600 nm, the angular width of the fringe formed on a distant screen is 0.1° . Find the spacing between the two slits.

11. Two Polaroids P_1 and P_2 are placed with their pass axes perpendicular to each other. Unpolarised light of intensity I_0 is incident on P_1 . A third polaroid P_3 is kept in between P_1 and P_2 such that its pass axis makes an angle of 60° with that of P_1 . Determine the intensity of light transmitted through P_1 , P_2 and P_3 .

12. Two wavelengths of sodium light of 590 nm and 596 nm are used in turn to study the diffraction taking place at a single slit of aperture 2×10^{-6} m. The distance between the slit and the screen is 1.5 m. Calculate the separation between the positions of first maxima of the diffraction pattern obtained in the two cases.

13. The ratio of the intensities at minima to the maxima in the Young's double slit experiment is 9:25. Find the ratio of the widths of the two slits.

14. In Young's double slit experiment, using monochromatic light L_1 of wavelength 700 nm, 10th bright fringe was obtained at a certain point P. If monochromatic light of wavelength 500 nm is used in place of L_1 , then which bright fringe will be obtained at the same point P?

15. In Young's double slit experiment using monochromatic light of wavelength 600 nm, 5th bright fringe is at a distance of 0.48 mm from the centre of the pattern. If the screen is at a distance of 80 cm from the plane of the two slits, calculate: (i) distance between the slits (ii) fringe width.

16. Monochromatic light of wavelength 650 nm falls normally on a slit of width 1.3×10^{-4} cm and the resulting Fraunhofer diffraction is obtained on a screen. Find the angular width of the central maxima.

PRACTICE PROBLEMS

1. In Young's double slit experiment, 2 narrow parallel slits 0.1 mm apart are illuminated with monochromatic light of wavelength 5890 Å, to get fringe pattern on a screen 1m away from the slits. Calculate the distance of (i) 5th dark fringe from the central bright fringe (ii) 3rd bright fringe from the central fringe.
2. In Young's double slit experiment, the distance between 2 coherent sources is 3 mm and the screen is 0.9 m from the sources. The 15th dark fringe is at a distance of 2 mm from the central bright fringe. Find: (i) wavelength of light (ii) shift of the 10th bright fringe when the separation between screen and slits is decreased by 0.1 m.
3. How does the angular separation of interference fringes change, in Young's experiment, if the distance between the slits is increased?
4. A beam of light, consisting of two wavelengths 560 nm and 420 nm, is used to obtain interference fringes in a Young's double slit experiment. Find the least distance from the central maximum, where the bright fringes, due to both the wavelengths coincide. The distance between the two slits is 4.0 mm and the screen is at a distance of 1.0 m from the slits.
5. What is the effect on the interference fringes to a Young's double slit experiment when (a) the separation between the two slits is decreased? (b) The width of the source-slit is increased? (c) The monochromatic source is replaced by a source of white light? Justify your answer in each case.
6. The intensity at the central maxima in Young's double slit experimental setup is I_0 . Show that the intensity at a point where the path difference is $\lambda/3$, is $I_0/4$.
7. Unpolarised light of intensity I is passed through a Polaroid. What is the intensity of the light transmitted by the Polaroid?
8. Yellow light ($\lambda = 6000 \text{ Å}$) illuminates a single slit of width $1 \times 10^{-4} \text{ m}$. Calculate the distance between two dark lines on either side to the central maximum, when the diffraction pattern is viewed on a screen kept 1.5 m away from the slit.
9. between two Polaroids's placed in crossed position, a third Polaroid is introduced. The axis of the third Polaroid makes an angle of 30° with the axis of the first Polaroid. Find intensity of transmitted light from the system assuming I_0 to be the intensity of polarised light obtained from the first Polaroid.
10. The distance between the first and the sixth bright fringes formed in Young's double slit experiment is found to be 12.5 mm. The separation between the slits is 0.12 mm and the distance of the screen from the slits is 60 cm. Calculate: (i) the wavelength of the light used (in nm) (ii) the fringe width.
11. Light of wavelength 600 nm falls normally on a slit of width 1.2 mm producing Fraunhofer diffraction pattern on a screen. Calculate the angular position of the first minimum and the angular width of the central maximum.

12. Green light is incident at the polarizing angle on a certain glass plate. The angle of refraction is 32° . What are: (i) the polarizing angle (ii) the refractive index of glass?
13. Determine the angular separation between the central maximum and the first order maximum of the diffraction pattern due to a single slit of width 0.25 mm when light of wavelength $5890 \text{ \AA}$ is incident on it normally.
14. Find Brewster's angle for water having refractive index 1.33.
15. A parallel beam of monochromatic light of wavelength 500 nm is incident normally on a rectangular slit. The angular width of the central bright fringe of Fraunhofer diffraction is found to be 60° . Find the width of the slit in metre.
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